

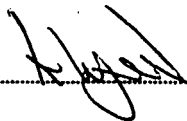
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Business Process Document

Specification for installation of cable routes forming part of the traction distribution system

Endorsement & Authorisation

Endorsed by:



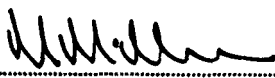
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2	April 2006	Re-issued as a SAF3 Business Process Document

Implementation

1 April 2006

Compliance

This Network Rail specification is mandatory and must be complied with by Network Rail and its contractors from 1 April 2006

When this specification is implemented, it is permissible for all projects, which have formally completed GRIP Level 4 to continue to comply with the Issue of any relevant Network Rail standards current when GRIP Level 4 was reached and not to comply with requirements contained herein unless the head of electrification and plant has stipulated otherwise in the accompanying briefing note.

Where the head of electrification and plant has specified that all projects will comply with the requirements, specific authority not to comply can be sought in the usual manner using the appropriate non-compliance authority processes described in Network Rail standard, [NR/SP/STP/045](#) *Managing the process for non-compliance authority* (formerly RT/LS/P/045).

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I Purpose

This document details the requirements for the design, refurbishment and construction of new cable routes and the refurbishment of existing cable routes for high voltage a.c. power distribution cables and associated pilot supervisory cables, signalling supply distribution and point heater cables, a.c. and d.c. traction cables and other cables used on electrical distribution systems.

2 Scope

This specification states the requirements for the design, refurbishment and construction of cable routes on electrical distribution systems.

3 Reference documentation

RT/CE/S/039	Network Rail Company Specification – Protective treatments for Network Rail infrastructure
RT/CE/S/044	Network Rail Company Specification – Undertrack crossings
NR/PS/ELP/00008	Network Rail Company Specification – High voltage cables and accessories for traction supplies
NR/PS/ELP/00013	Network Rail Company Specification – Cable troughing
RT/LS/CAT/004	Network Rail Company Specification – Signalling standard drawings
NR/LS/S/011	Network Rail Company Specification – Prevention of damage to and danger from surface and buried services
EN ISO 9001	Model for quality assurance in design, development, production, installation and servicing
EN 124	Gully tops and manhole tops for vehicular and pedestrian areas. Design requirements, type testing, marking, quality control
BS 7671	Requirements for electrical installations: IEE Wiring Regulations

4 Definitions

Approval	Prior approval or acceptance, as appropriate, by Network Rail, or other where specifically designated in the text.
Bellmouth	Transition section between surface trough route and duct route.
British Standard	A standard published by the British Standards Institution. Its alphanumeric identity is prefixed by BS or BS EN.
Equipment	Equipment including all component parts and accessories.

European Standard	A standard published by the European Committee for Standardisation or by the European Committee for Electrotechnical Standardisation. Its alphanumeric identity is prefixed by EN.
International Standard	A standard published by the International Electrotechnical Commission. Its alphanumeric identity is prefixed by IEC.
Network Rail Company Standard	A collective term for a Network Rail Product Specification, Network Rail Procedure or a Network Rail Guidance Note, published by a directorate of Network Rail.
Procurement Specification	The contract specific Network Rail documentation, which, together with product specification(s) forms part of an Invitation to Tender or Technical Workslope.
Traction cables	25 kV ac and 750/660 V d.c. positive and negative cables for distribution of traction supplies from substations and along the track

5 General clauses

5.1 Safety, legislation and regulations

Measures shall be taken to ensure the safety of persons concerned in the installation, operation, maintenance, repair and its eventual disposal of cable route.

At all times the cable installation shall comply with the requirements of all relevant statutory acts, rules, regulations or orders.

5.2 Design, manufacture and workmanship

Materials used shall be designed and manufactured in accordance with best practice, taking cognisance of the system in which it will operate. All material shall be of the quality and of the type most suitable for working under the conditions specified for the full service life of the cable route.

5.3 General

The cable route shall be designed, constructed and refurbished so as to be compatible in every respect with the required performance of the cables and shall be fit for its intended purpose when the cables are connected into the systems described in clause 5 and as may be supplemented in the procurement specification.

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The cable route shall also provide adequate mechanical and fire protection to the cables and shall be designed to ensure that the minimum bending radius of the cables is not compromised. It shall be the most cost effective design, taking into consideration the type of cable to be installed, the topography of the site, the trackside environment, the method of installation and the cost of materials.

5.4 Environmental, service and installation conditions

The completed cable route system shall be suitable for the rated characteristics of the cables to be installed or already installed.

The service conditions are:

- a) the minimum ambient air temperature shall be -15 °C;
- b) the maximum ambient air temperature shall be +40 °C;
- c) the cable route is subject to step heating from direct sunlight and is located in areas where trackside fires are likely;
- d) the ambient air is polluted by the presence of airborne iron dust from train brakes;
- e) the cable route is subject to vibration caused by passing trains;
- f) the cable route is subject to physical damage from trackside works;
- g) the cable route will be exposed to electromagnetic interference expected on the railway infrastructure;
- h) the cable route may be installed in areas electrified at 25 kV a.c., 1500 V d.c., or 750 V d.c.

5.5 Maintainability

The cable route shall be designed for minimum maintenance during its service life. Individual components of the route shall be readily replaceable with minimum effect on the installed cables.

5.6 Reliability and availability

The electrical power distribution system will normally remain continuously energised and in service to provide power supplies to traction and signalling equipment. Failure can cause interruption of the operation of the railway; hence a high level of reliability of the cable route to protect the installed cables is required.

5.7 Service life

The cable route material shall be capable of complying with this specification, including performing its intended purpose, for at least thirty years or as specified in the procurement specification, without deterioration in reliability. The cable route shall be so installed and adequately supported to ensure that under normal ground conditions it is retained in its design position for its service life (at least thirty years or as specified in the procurement specification).

5.8 Resistance to mechanical and environmental damage

The cable route shall be robust and shall provide the cables throughout their length with a high degree of protection from mechanical, abrasive, fire and other environmental damage.

5.9 Bending and other mechanical stresses

The bending radius of the cable route shall not be less than that recommended by the cable manufacturer. Any other stresses imposed on the cable by the design and construction of the cable route shall not exceed the cable manufacturer's recommendations. Wherever practicable, all cables shall be continuously supported throughout the route.

Particular consideration shall be given to the movement of the cable within the route due to thermal expansion and contraction of the cable during ambient temperature variations and cyclic loading conditions.

Cable routes using intermittent cleated supports will be accepted where continuous support cannot be achieved such as vertical runs or limited clearance situations. Such installations shall be designed to ensure that the stresses imposed on the cable do not exceed the cable manufacturer's recommendations. The spacing between support cleats and the cable sag between supports shall be in accordance with the cable manufacturer's recommendations.

5.10 Interfaces with existing cable systems

New cable routes, or extensions to existing cable routes shall ensure that the requirements of this specification are totally maintained at interfaces with existing cable route installations.

5.11 Interchangeability

All like equipment shall be interchangeable as a whole and also as regards its parts. Any such change or replacement shall not reduce the capability of the cable route to conform to the requirements of this specification.

5.12 Finish and resistance to corrosion

Metal components, accessories, fittings and fastenings (e.g. nuts, bolts, washers etc.) shall be protected against corrosion, including galvanic corrosion of dissimilar materials, for the life of the cable route. All ferrous material shall be galvanised or manufactured from stainless steel. Protective finishes shall comply with RT/CE/S/039 as appropriate. Procedures for preparation and applying protective finishes shall be submitted for approval.

6 Drawings and documentation

The following drawings and other documentation shall be provided by the contractor for approval:

6.1 Cable route diagrams

Cable route diagrams scale 1/1250 based on Ordinance Survey Mapping, with reference given to any relevant buried cable sections, and showing:

- a) mile post and/or kilometre posts;
- b) measurements in metres from the route start;
- c) type of route laid over various sections of the route;
- d) details of cables laid over the various sections of the route;
- e) details of existing route(s) in near vicinity and the methods of crossing them when applicable;
- f) joint positions with measurements to nearest mile post and/or kilometre post;
- g) cross section of cable route showing cable type and arrangement;

Note: Cross section needs to be shown wherever there is a change of route parameters e.g., trenching size change or presence of an undertrack crossing.

- h) cable drum or batch numbers and cable manufacturer;
- i) date of work.

6.2 Buried cable charts

Buried cable charts or drawings for buried sections of the route and for road and track crossings showing:

- a) position of the route with respect to fixed locations and cross sections indicating the type of route;
- b) number of ducts in the crossing;
- c) position of the new cable;
- d) redundant cable within the route;
- e) depth and type of backfill used.

- f) contents of the route, e.g. type and sizes of cables, troughing material/size/reference details.

6.3 Other drawings

Drawings of sections of the route where special support arrangements are used with detailed drawings of supporting steelwork, cleating arrangements etc.

6.4 Record drawings

All drawings shall be provided as hard copy and in electronic format suitable for Microstation or as specified in the engineering specification.

7 Technical requirements for cable route materials

Ducts and troughing, and other cable route components and accessories shall provide adequate robustness to withstand the mechanical stresses experienced during installation and all service conditions, including during short circuits on the cables contained within the route.

The cable route components shall be of an interlocking design and shall provide a continuous smooth internal surface for the cable. Covers and lids shall be a close fit and positively located using registration to prevent inadvertent displacement after installation.

7.1 Reinforced concrete troughing and lids

The troughing and associated materials shall comply with the requirements of Network Rail specification [NR/PS/ELP/00013](#) and shall be supplied in lengths not exceeding 1000 mm.

7.2 Troughing manufactured in alternative materials

Troughing shall comply with the requirements of Network Rail specification [NR/PS/ELP/00013](#) and may be supplied in lengths up to 3000 mm dependant upon weight and strength.

Metallic troughing shall not be used except for approved exceptional circumstances.

Troughing manufactured from glass reinforced polyester or from heavy duty polyvinyl chloride (PVC) will be acceptable subject to meeting the material requirements of Network Rail specification [NR/PS/ELP/00013](#). In particular, these materials shall comply with the fire resistance requirements and shall have satisfactorily completed the fire resistance test laid down in Network Rail specification [NR/PS/ELP/00013](#).

7.3 Ducts for under road and under track crossings

Ducts shall normally be of the reinforced flexible type but other materials as specified in Network Rail specification [RT/CE/S/044](#) may also be used. Metallic ducts shall not be used.

7.4 Tubes for use in routes through station platforms

Heavy duty rigid polyvinyl chloride tubes or preformed duct blocks shall be used through station platforms. These shall be to an acceptable design.

7.5 Buried cable route and joint markers

Each marker shall consist of a label engraved yellow laminate or similar approved material with black lettering securely fitted to a concrete marker, adjacent wall or building with non removable fixings as required by Network Rail specification [NR/LS/S/011](#).

Monumental or block concrete cable route and joint markers shall be designed such that the label is supported and protected against long term damage.

8 Route design requirements

Concrete cable troughing shall be set into the ground for a minimum depth of 100 mm, and where practicable, at a minimum distance of 3.0 m from the gauge face of the nearest running rail.

At obstructions or where topographic features prevent this distance being achieved, the cable troughing shall be installed at the maximum practicable distance but not less than 1410 mm from the gauge face of the nearest running rail.

Routes shall be designed so that the minimum bending radius of the largest cable to be installed in the route is not infringed.

Route capacity shall comply with [BS 7671](#), where applicable.

For 25kV electrification, 2 track feeder cables per route can be designed.

For dc electrification, 2 dc track feeder cables per route can be designed and installed.

Where possible, a maximum gradient of 1 in 5 shall not be exceeded except in duct/tube routes and locations where the cables are to be positively fixed by cleats or other approved means. At locations where there are changes of gradient, special arrangements shall be applied to ensure that the cables are not subject to mechanical stress or are damaged by movement.

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Cable troughing shall not be installed within authorised walking routes, unless it is unavoidable, when special protective methods to ensure a safe walking surface shall be adopted.

Positioning cable routes shall avoid compromising access to trackside or off-track drainage and shall not be placed directly above access chambers or catch pits which hinder drain inspection and maintenance.

8.1 Type of route to be used

Surface laid reinforced concrete troughing shall be used, wherever practicable, and shall be in accordance with the BRS-SC series of drawings listed in [NR/LS/CAT/004](#). The preferred size shall be C/1/9 and for larger sizes C/1/19 and C/1/43, which shall be used with approved reduction sections.

Troughing manufactured of alternative materials such as glass reinforced polyester or polyvinyl chloride shall be used where ground conditions or clearances are not suitable for surface laid reinforced concrete troughing (e.g. unstable embankments, on embankments where the cess is narrow, situations of limited clearance). Such troughing shall normally be installed above ground on posts or attached to walls and adequately supported at appropriate intervals.

Where use of surface or aboveground troughing is not practicable, lengths of buried concrete troughing, duct lines, cables buried direct or cables fixed in cleats may be used.

8.2 Routing of high voltage cables

In areas where two high voltage cable circuits follow the same railway line, separate routes shall be provided unless space limitations necessitate the use of dual troughing. Diverse routing shall be achieved, wherever practicable, by locating the two routes on opposite sides of the track. For 25 kV electrification, 2 track feeder cables per route can be installed.

Cable routes for high voltage cables shall be separated by a minimum of 1 m and shall, wherever practicable, be routed on the opposite side of the track from cable routes for signalling and/or telecommunications cables. Where routing on opposite sides of the track is not possible, high voltage and other cable routes shall be separated by a minimum of 1 m and the length of parallelism shall be kept to a minimum. Parallelism shall not exceed 50 m.

8.3 Cable route crossings of roads and running lines

Where cable routes cross roads and running lines, buried duct lines shall be provided comprising a minimum of four ducts. In addition, spare ducts shall be provided in accordance with the requirements of Network Rail specification [RT/CE/S/044](#).

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Under track crossings shall be designed to comply with requirements of Network Rail specification [RT/CE/S/044](#) except that reinforced flexible ducts shall be used.

For typical cable route crossings under road and running lines see Drawing No A437/00/CV/40572, Appendix A.

Existing underline or overline bridges or gantry type cable bridges shall only be used where approved and an under track crossing is costly or cannot be provided.

Under road crossings shall be designed to comply with the standards laid down by the highway authorities.

8.4 Cable routes in station platforms

Cable routes buried in station platforms shall consist of a minimum of four approved heavy duty smooth bore rigid polyvinyl chloride tubes or duct blocks encased in concrete or other approved fill material for each high voltage cable and its associated pilot cable. One tube per cable shall be used. Spare ducts in accordance with the formula in [NR/SP/CE/044](#) shall be provided.

8.5 Other buried cable routes on Network Rail infrastructure

Buried cable routes on Network Rail infrastructure shall consist of either buried reinforced concrete troughing with double lids or duct/tubes enclosed in concrete. Where neither of these arrangements is practicable, the cables may be buried direct in the ground with acceptable protection from above.

8.6 Other buried cable routes off Network Rail infrastructure

Buried cable routes located outside Network Rail infrastructure shall consist of either duct/tubes enclosed in concrete or shall be buried direct in the ground.

8.7 Duct/tube routes

Duct/tube routes shall be designed with the minimum number of joints. Where lengths are less than 30 m, a continuous duct without joints shall be used.

Where duct/tube routes are provided, separate ducts/tubes shall be provided for each high voltage cable or for each phase cable (where 3xsingle core cables are being used), for each d.c. positive and each d.c. negative cable and for up to two pilot/supervisory or low voltage cables subject to the diameter of each cable or the equivalent combined diameter of the two cables not exceeding 75mm.

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8.8 Temporary cable routes

Temporary cable routes shall comply with this specification in as far as it is applicable. Reliability of service and safety shall not be compromised in comparison with a permanent cable route.

9 Construction requirements

9.1 Route survey

The route shall be surveyed and route notes produced for approval before starting ground preparation work. The survey shall use photographic, video and other available data to minimise the number of trackside visits. All relevant information held by Network Rail will be provided to the contractor including buried services details and relevant sections of the Territory Hazard Directory.

The route notes shall give details of the type of route to be installed and the existing route to be reused, measured in metres from the start of the route, shall reference mile/kilometre posts, locations of road and track crossings and shall include details of any existing routes and other relevant trackside infrastructure such as station platforms, signal positions, bridges, etc. Where possible, provisional locations of cable joints shall be included.

The route survey shall identify all existing routes and their suitability for refurbishment and reuse, and present condition.

Before work starts at any location, a survey of the site by visual means and mandatory use of cable and pipe locating equipment shall be carried out in accordance with Network Rail specification [RT/LS/S/011](#).

9.2 Route construction

The cable route shall be installed to form a continuous smooth internal surface. Troughing shall be installed with matching lids and lids shall be offset by a minimum of 100mm to overlap the troughing joints.

9.3 Surface reinforced concrete troughing

Surface troughing shall normally be installed straight and level with interlocking ends butting and shall be installed on a firm, even base and cut into the ground to a minimum of 100 mm of their depth. Where ground conditions are such that the minimum cut in depth would not adequately support the route, the cut in depth shall be increased accordingly.

Where troughing is installed adjacent to walking routes, permanent way huts, signal boxes and other such trackside locations, it shall be buried such that the top of the trough lids are level with the surrounding ground surface.

Troughing made of other materials shall not be installed at ground level, except where approved.

9.4 Buried reinforced concrete troughing

The troughing shall be provided with double lids and installed with lids offset to well overlap the troughing joints. Cable marker tape incorporating a metalised strip shall be installed between 150mm and 250mm below ground level.

At occupational and accommodation crossings or roadways not covered under the highway regulations where vehicular traffic cross the cable route, it shall be buried to a depth of 750 mm below ground level. At other locations, the depth may be reduced to 600 mm below ground level or 500 mm below the bottom of sleepers when crossing sidings or behind buffer stops.

At catch points, the route shall be:

- a) buried for a minimum length of 50 m beyond the tips of the points in the normal direction of rail traffic; or,
- b) positioned well clear of the risk of damage arising from an intentionally derailed rail vehicle for a minimum length of 50m beyond the tips of the points in the normal direction of rail traffic.

9.5 Reinforced concrete troughing at bridges and in tunnels

At underline and overline bridges and in tunnels where the distance between the gauge face of the nearest running rail and the abutment/tunnel wall or bridge girder is greater than 1250 mm, providing that it does not interfere with the ballast shoulder, troughing may be installed in the cess subject to line speed, required ballast shoulder and approval of Territory Track Engineer. In such circumstances, the troughing shall be buried to lid level and shall be located close to the abutment/tunnel wall or bridge girder. In tunnels and other locations where refuges are provided, the cable route shall allow unrestrictive access to the refuges and shall not cause a tripping hazard.

9.6 Cable route crossing under roads and running lines

Cable ducts shall, where practicable, be of one continuous length from end to end. They shall normally be located at right angles to the road or running lines and curved at each end to join up with the trackside cable route. The bends shall be as large as possible and shall not be less than the cable manufacturer's recommended minimum bending radius for the largest cable to be installed.

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The ducts shall be graded to the surface whilst maintaining the required cover under the road or the sleepers of the running lines. A maximum gradient of 1 in 5 shall be provided unless site restrictions make this impossible. At each end of the duct line, bellmouths with its lids at surface level shall be provided to connect the duct line with the troughing route. Alternatively, where site arrangements dictate, cable pits may be provided.

The ducts shall be separated with a minimum clear distance of 150 mm between each duct, encased in concrete, or compacted granular fill media compacted with a mechanical aid, with a minimum of 150 mm of concrete, or granular fill media, cover above and below the ducts. The concrete surround shall extend under the road and associated pavement and under the track up to the bellmouths or cable pit or for a distance of 2000 mm beyond the sleeper ends whichever is the lesser.

Cable marker tape incorporating a metalised strip shall be installed between 150mm and 250mm below ground level.

Between high voltage cable ducts and low voltage (including signalling and telecommunication) cable ducts, minimum separation of 300mm shall be provided.

The minimum depth of the ducts under the road shall be 750 mm. Reinstatement after installation shall comply with the local authority and highway requirements. For typical examples of reinstatement after installation see Drawing No A437/00/CV/40559, Appendix A.

The minimum depth and construction of duct lines under the running lines shall comply with the requirements of Network Rail specification [RT/CE/S/044](#).

For typical cable route crossings under road and running lines see Drawing No A437/00/CV/40572, Appendix A.

Following installation, the ducts shall be cleaned, roped with visible protruding ropes, and sealed with a removable foam sealant to approximately 100 mm penetration into the duct to prevent ingress of stones.

When digging undertrack crossings, the ballast shall be protected from contamination by spoil using sheets or boards etc.

9.7 Cable routes along roads

Cable routes running along roads shall comply with the local authority and highways requirements.

If the route runs in the road or under a tarmac pavement, the route shall consist of either ducts encased in concrete to the same standard as routes crossing roads (see 9.6 above) or shall be buried direct (see 9.10 below).

Cable marker tape incorporating a metalised strip shall be installed between 150mm and 250mm below ground level.

Where the route can be located in ground alongside the road or pavement, the cable may be buried direct subject to approval.

9.8 Cable routes through station platforms

Excavations shall be suitably protected by the use of covers, barriers, tapes, warning notices etc. and all excavations shall be covered during train service times. After completion the platform shall be re-instated at least to existing standard.

The tubes/duct block shall be buried with at least 125mm cover to the platform surface, shall be separated with a minimum distance of 50 mm between each duct and shall be surrounded by a minimum of 50 mm of concrete. Cable marker tape incorporating a metalised strip shall be installed between 150mm and 250mm below ground level.

Typical arrangement is as shown on Drawing No A437/00/CV/40558, Appendix A

The tubes/duct block shall be cleaned, roped and sealed following installation.

Cable routes running trackside at stations between platform and running line shall be provided with loops at end of platforms or other suitable locations to allow lifting of cables onto the platform when ballast cleaning or relaying new ballast.

9.8.1 Changes in direction of cable routes in station platforms

At major changes in direction, and at buried transitions between routes of different types as may be required. Cable pits of approved design shall be provided. Steel pit covers in accordance with [EN 124](#) shall be provided for cable pits as follows:

- a) In areas of platform used only for pedestrian and hand propelled barrow traffic;
- b) In areas of platform which may be used by motorised vehicles.

9.9 Trough route installed above ground level

Glass reinforced polyester or heavy duty polyvinyl chloride troughing shall be used for above ground routes supported on brackets, typically as shown on Drawing No A437/00/CV/40543, Appendix A, attached to lineside structures or, at 1.5 m intervals, to reinforced concrete posts, to Drawing No A437/00/CV/40552, Appendix A. Metal posts can be used as an alternative.

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The maximum height to the underside of the troughing shall be 500 mm above ground level. The route shall be located at a minimum distance of 3 m from the running rail. Where this clearance is not practicable, cable troughing shall be installed at the maximum practicable distance, but not less than 1410 mm from the rail.

At bridges, in tunnels and other restricted clearance locations, an above ground route may be used provided that the minimum gauge clearance is maintained.

Where applicable, lids shall be retained by stainless steel straps which shall not damage the troughing units unless otherwise approved.

The posts shall be set into the ground with a minimum of 150 mm of concrete around the post for its entire buried depth. The posts shall be installed to a depth at least equal to the above ground height. The length of the posts shall be increased to penetrate below the slip circle depth at locations of known or obvious susceptibility to landslip.

The troughing shall be supported on metal hangers that shall be fixed securely to posts, walls, bridges or other lineside structures, typically as shown on Drawing No A437/00/CV/40543, Appendix A.

The method of attachment and installation procedure to be used on walls, at bridges and at other lineside structures shall meet the requirements of the Territory Civil Engineer.

9.10 Cables buried direct in the ground

The cable trench shall be prepared as far as practicable to avoid damage to the cable during installation and in service. The base of the trench shall be smooth and filled with a layer of sand. The cable when installed in the trench shall be set level and shall be surrounded with a minimum of 150 mm of sand.

Unless otherwise required to satisfy the requirements of a Statutory Authority for sections of the cable route crossing their property, the minimum cover for high voltage cables buried directly in ground shall be 750 mm below finished normal ground level.

All cables shall be protected from possible disturbance, abrasion or mechanical damage when entering the ground or transferring from a lower depth into cable troughing by the use of ducts or other equivalent means.

The presence of the cable shall be given by a cable marker tape/mesh laid in the cable trench 150-250 mm below finished ground level. This tape/mesh shall be yellow approximately 300 mm wide and marked at regularly with a black electrification danger sign and shall incorporate a detector mesh to ensure identification by cable and pipe locating equipment. Cable tiles shall be provided where required by the engineering specification.

9.11 Buried cable route and joint markers

Buried cable routes and duct/tube routes shall be marked with cable markers at intervals of 20 m and at any change in route direction. The markers shall show the information required by Network Rail specification [RT/LS/S/011](#) and shall include the cable designation. In the cess and near walking routes, the cable markers shall be of a type and located such that they are not a tripping hazard.

Cable joint markers shall be provided adjacent to the cable route to indicate each joint position and shall include the feeder description, the joint number and where applicable, the phase colour. These markers shall not create a tripping hazard.

9.12 Cables routes cleated or supported from structures, walls etc

This method shall only be used where alternative types of route construction are not practicable and at locations where the cable route will be secure from physical damage, i.e. clear of pedestrian and vehicular traffic, and does not itself present danger to personnel or equipment. Any work, including drilling or attaching fixtures etc., shall meet civil engineering specifications.

The cables shall be provided with a protective covering to provide security, fire protection and to reduce heating due to radiation from the sun. Special consideration of the risk and effect of fire shall be made on vertical cable runs and appropriate protective measures applied.

The cables shall be secured in such a way as to prevent undue mechanical stresses and movement under short circuit conditions. All cable cleats to walls, structures etc., shall be installed at the spacing required to produce the level of cable sag between supports in accordance with the cable manufacturers' recommendations.

9.13 Crossings of cable routes

Where cable routes cross the following shall apply, wherever practicable:

- a) d.c. traction cables shall be routed underneath the main lineside high voltage or signalling/telecommunications cable route. Where the lineside cable route is elevated, the d.c. traction cables shall be laid at surface level. Where the main cable route is at surface level, the d.c. traction cables shall be buried and routed underneath using ducting which facilitates rapid replacement of the d.c. traction cables;
- b) as an exception to (a), where the main cable route is buried, the d.c. traction cables may be routed over the top at surface level;
- c) a minimum 75 mm air clearance between d.c. traction cables and other cable routes shall normally be maintained;

- d) where the 75 mm clearance between d.c. traction cables and a main cable route is encroached the d.c. traction cables shall be provided with an approved type of additional mechanical and fire protection;
- e) where a.c. traction cables cross ST cable routes, the ST route shall pass underneath the high voltage cable route. Where other arrangements are proposed these shall be approved before work commences.
- f) joints in a.c. or d.c. traction cables shall not be installed within 1000 mm of any cable route crossing.

9.14 Provision for jointing

Joint bays shall be provided as required, e.g., where there is a transition from oil filled to solid insulation cable or where cable joints cannot be accommodated within the standard C-I-9 cable troughing.

Where cable routes of different types meet, transition arrangements shall be provided that maintain the protection and installation requirements of this cable route specification. Typically transition arrangements are shown on Drawing No A437/00/CV/40520, Appendix A

9.15 Route requirements for installation of single core cables

Single core high voltage a.c. distribution cables on three phase circuits shall be grouped in close trefoil formation and strapped at 2 metre intervals to ensure that under maximum fault conditions the cables are securely held in position. Where installation conditions make the use trefoil cleats impossible, the single core cables may be independently cleated up to a maximum length of 25m. The straps shall be non-metallic and shall not cause damage to the cables or other cables in the route.

10 Refurbishment and reuse of existing cable routes

Existing cable routes may be refurbished and reused where practicable. Re-used route materials shall be capable of complying with the requirements of this specification and, where appropriate, [NR/PS/ELP/00013](#), for the specified service life.

Reinforced cable troughing and lids (including concrete joint bay units) of imperial size may be reused where the troughs are in sound condition and the route will provide the cable with adequate mechanical and fire protection and the cable shall not be damaged by movement within the route.

Internal rough surfaces shall be made smooth. Sand in existing sand filled troughing or ducts shall be removed.

Ref:	NR/SP/ELP/27224
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Date:	April 2006

Any concrete troughing and lids with reinforcement showing or sufficiently broken corners/edges to allow ballast to penetrate the route shall not be reused. All longitudinally cracked troughing lids shall be replaced.

Duct routes under roads and the running lines may be reused subject to sufficient spare ducts being available unless the ducts are constructed of an asbestos based material. Before use, the ducts shall be rodded to ensure that no blockage or discontinuities exist. Following installation and commissioning of the new cable, the existing cables shall be removed to provide the spare ducts unless specifically approved.

Existing duct routes in station platforms and all buried cable troughs may be reused. Other above ground routes and/or supports may also be reused.

Existing sections of non standard route shall be removed and a new route installed. Sections of route comprising asbestos containing materials shall not be reused.

11 Removal of redundant routes and cables

Except for buried ducts/tubes, wherever practicable, all redundant route material shall be dismantled and removed from site. Where removal may disturb or collapse the ballast shoulder, the route shall be left, troughing lids shall be removed, troughing back filled with uncontaminated ballast and a record of these locations shall be kept in the Health & Safety File.

Where possible existing posts and their foundations shall be completely removed. Where this is not practical, the posts shall be broken at least 150 mm below ground level.

All redundant cable shall be cut up and removed from site. Cables shall be removed, where reasonably practicable, from buried sections of the route. Any oil filled pressure cable or resin insulated cable left on site shall be drained, flushed through and capped to prevent leakage of the insulation materials.

Where buried cables are not removed, they shall be shown on buried cable charts and recorded in the Health and Safety file.

Ref:	NR/SP/ELP/27224
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Date:	April 2006

Appendix A Drawings

<u>A437/00/CV/40520</u>	Concrete ramp to accommodate transition unit
<u>A437/00/CV/40543</u>	General Arrangements of brackets on bridges and walls to support HV & Pilot cables and DC Traction cables
<u>A437/00/CV/40552</u>	Concrete steel reinforced post for use with glass fibre reinforced troughing
<u>A437/00/CV/40558</u>	Arrangement of UPVC tube route for h.v. and pilot cables through platforms
<u>A437/00/CV/40559</u>	Guide to types of flexible road construction for reinstatement of trenches in carriageways
<u>A437/00/CV/40572</u>	Typical arrangement of metric duct lines for track and road crossings

NOTES

* DIMENSIONS SUBJECT TO SITE CONDITIONS
ALL DIMENSIONS, UNLESS INDICATED OTHERWISE,
ARE IN mm
REDRAWN FROM SOUTHERN REGION DRAWING SR/A2/9/9

C01	23/11/04	ISSUED FOR CONSTRUCTION				MA	AA	AA
Rev.	Date	Description of Revisions				Drawn	Chkd	Appr.
Project Manager:		Network Rail Project Manager:				Network Rail Applications Manager:		

CONSTRUCTION ISSUE



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 Tel: 0207 620 3696
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Alliance Partner/Contractor

Alliance Partner/Contractor

Project Title

CONCRETE RAMP TO ACCOMMODATE
TRANSITION FROM CONCRETE
C-1-9 TROUGH TO GRP TROUGH

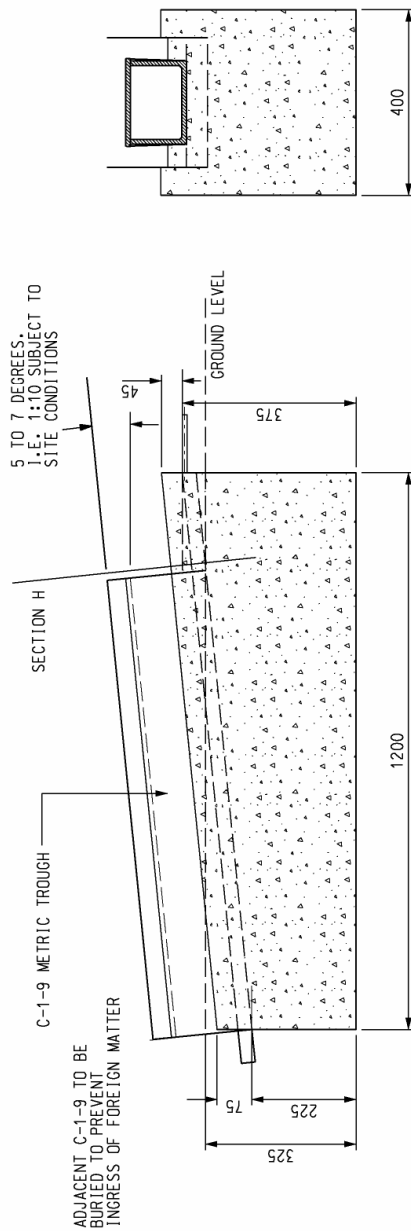
Designed		Checked	T. BENNETT
Approved	N. JORDAN	Responsible Engineer	N. JORDAN
Date	04/10/03	Drawn By	K. MEIN
Scale	1:100 @ A2	Sheet	4 of 4

CAD Plot. 1:10 @ A3 DESIGN/SOUTHERN REGION/CIVIL DRAWINGS

A437/00/CV/40520

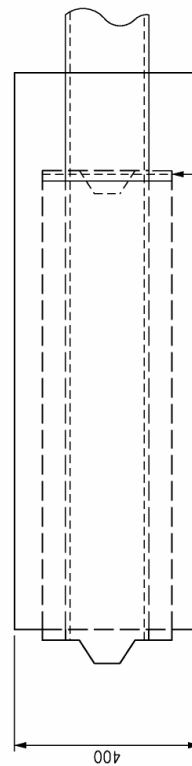
Size: A3

Ref:	NR/SP/ELP/27224
Issue:	2
Date:	April 2006



CONCRETE RAMP CAST INSITU

SECTION H-H



REMOVE SHARP EDGE OF G.R.P AND FILL BETWEEN EDGE OF G.R.P AND SIDE OF C-1-9 TO PREVENT CABLE DAMAGE. G.R.P TO BE BUTTED AND RAMPED C-1-9.

DO NOT SCALE

NOTES

STRAPPING FOR SECURING LID TO BE APPLIED ON THE BASIS OF ONE PER 2 METRE LENGTH.
GALVANISED COATINGS ON ITES 5, 6, AND 7 TO BE BS EN 1461

1. MILD STEEL X x 65 x 10 SUPPORT BRACKET, BS970 PT.1
2. MILD STEEL 345 x 65 x 10 SUPPORT BRACKET BS970 PT. 1
3. MILD STEEL X x 65 x 10 TIE SUPPORT BAR, BS970 PT1
4. GLASS FIBRE REINFORCED TROUGHING
5. GALVANISED 20 METRIC HEXAGON HEAD BOLT, BS4190 TABLE 11
6. GALVANISED 20 METRIC HEXAGON WASHER, BS4320 TABLE 3
7. GALVANISED 20 METRIC HEXAGON NUT, BS4190 TABLE 12
8. 12mm DIAMETER x 50mm LONG COACH BOLT WITH STANDARD WASHER, BS-1494 TABLE 15A

ALL DIMENSIONS, UNLESS INDICATED OTHERWISE, ARE IN mm

REDRAWN FROM SOUTHERN REGION DRAWING SR/A2/7722

CD	DT/DM	ISSUED FOR CONSTRUCTION	MA	AA	AA
Rev.	Date	Description of Revisions	Drawn	Chkd	Appr.
Project Manager	Network Rail Project Manager	Network Rail Applications Manager			

CONSTRUCTION ISSUE



Filing Bridge Court
2nd Floor
25 Abchurch Lane
London EC4N 3DF
Tel: 0207 493 3863
Fax: 0207 493 3862

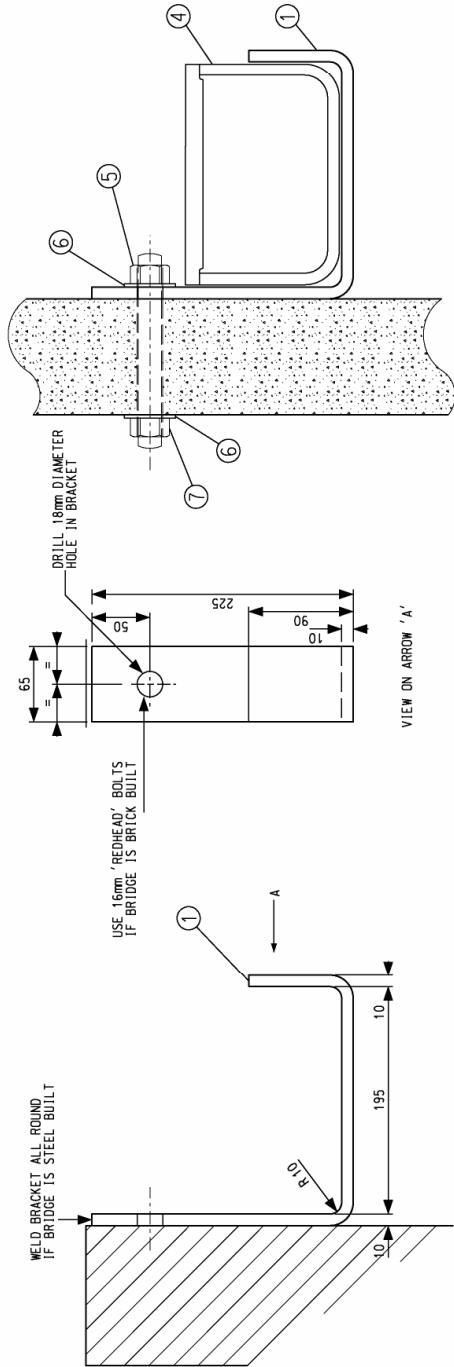
Network Rail Project Manager

Project Title

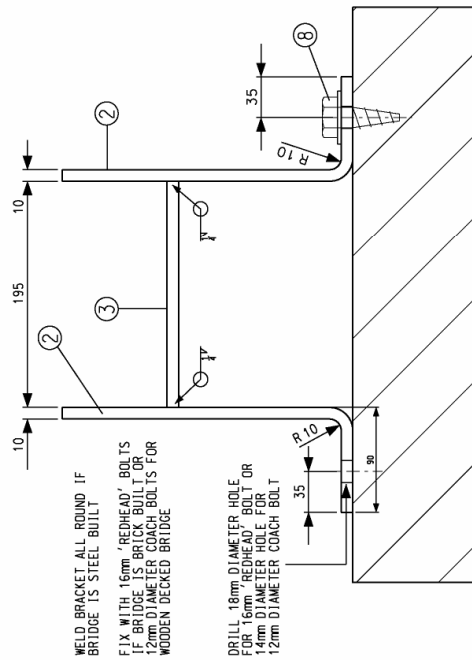
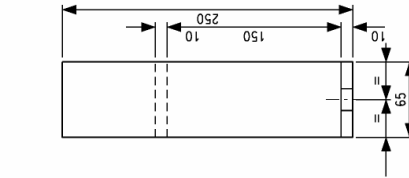
GENERAL ARRANGEMENTS OF BRACKETS ON BRIDGES AND WALLS TO SUPPORT HIGH VOLTAGE & PILOT CABLES & DC TRACTION CABLES

Designed	Checked	Approved	Authorised Engineer	N. JORDAN
Date	09/10/03	Drawn By	K. MEIN	
Scale	1:2 @ A3	Sheet	1 of 1	
CAD Ref.	...-DESIGN/SOUTHERN REGION/CIVIL DRAWINGS			
Drawing Number	A437/00/CV/40543			

Size: A3



ARRANGEMENT OF BRACKET FOR EITHER SIDE OF BRIDGE



ARRANGEMENT OF BRACKET ON TOP OF BRIDGE
ITEM NO.2 DEVELOPED LENGTH 345

All changes to this CAD generated drawing MUST be made via the master CAD file.

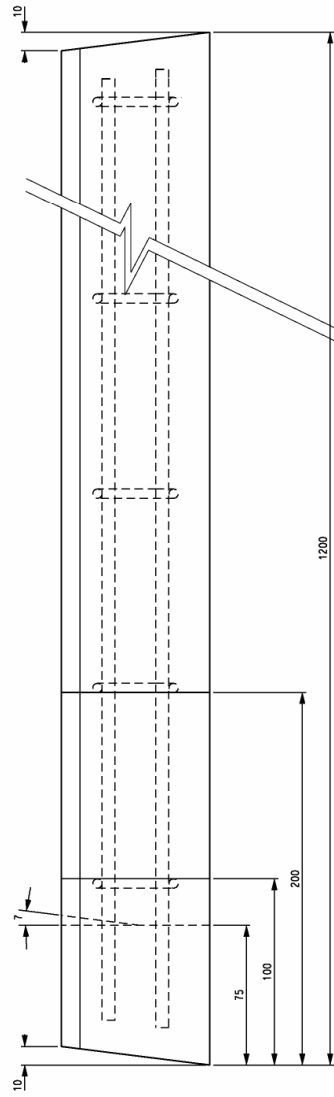
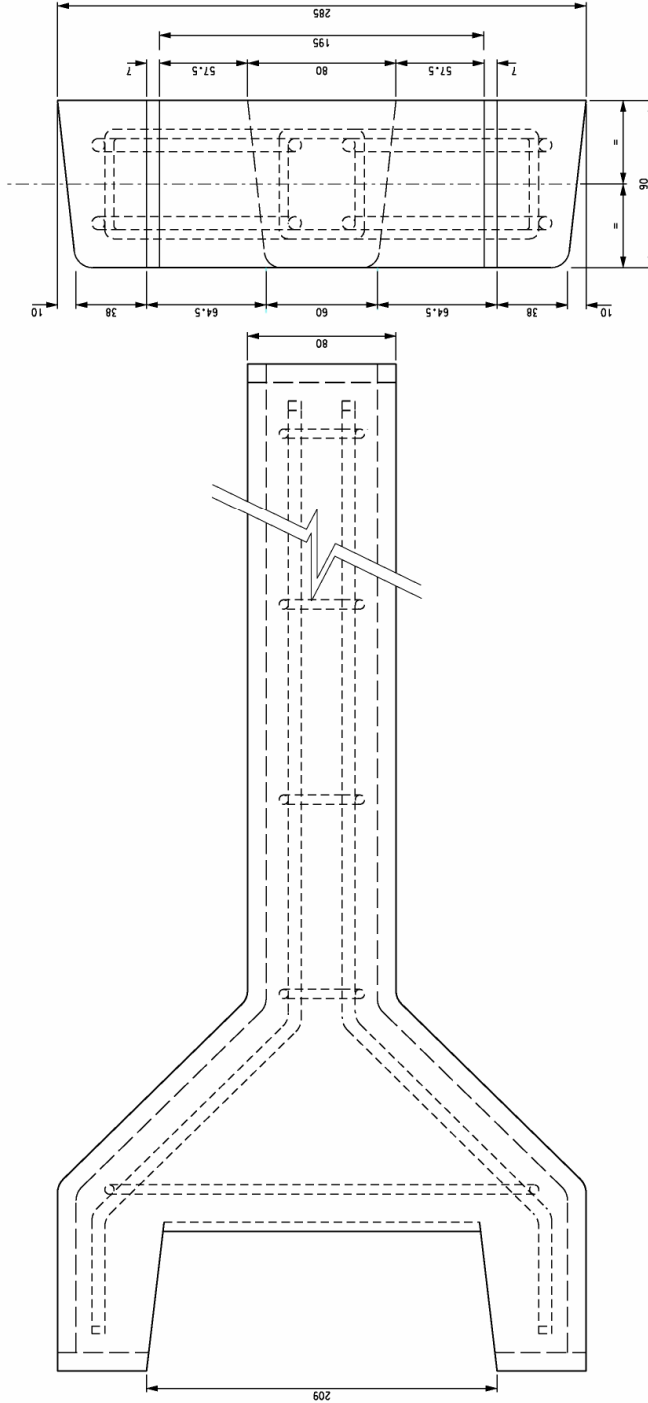
DO NOT SCALE

NOTES

DETAILS OF STEEL REINFORCING & ALL RADII
FOR CASTING PURPOSES TO BE DETERMINED BY
MANUFACTURER

ALL DIMENSIONS ARE IN MILLIMETRES

REDRAWN FROM SOUTHERN REGION DRAWING SR/A2/9846



COL	DATE	ISSUED FOR CONSTRUCTION	MA	AA	AA
Rev.	Date	Description of Revisions	Drawn	Chkd	Appr.
Project Manager		Network Rail/Project Manager	Network Rail/Project Manager		

CONSTRUCTION ISSUE



Filing Bridge Court
21-25 Blackwing Road
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Mobile: 0777 880 5832

Alfreda Thomas / Contractor

Ref:	NR/SP/ELP/27224
Issue:	2
Date:	April 2006
Project Title	CONCRETE STEEL REINFORCED POST FOR USE WITH GLASS FIBRE REINFORCED TROUGHING AND 'KENDUCT' PVC TYPE TROUGHING
Designed By	T. BENNETT
Approved	N. JORDAN
Responsible Engineer	N. JORDAN
Date	13/10/03
Drawn By	S. PANGK. MEIN
Scale	1:2.5 @ A3
Sheet	1 of 1
CAD Ref.	DESIGN/SOUTHERN REGION/CIVIL DRAWINGS
Drawing Number	A437/00/CV/40552
Revision	C01

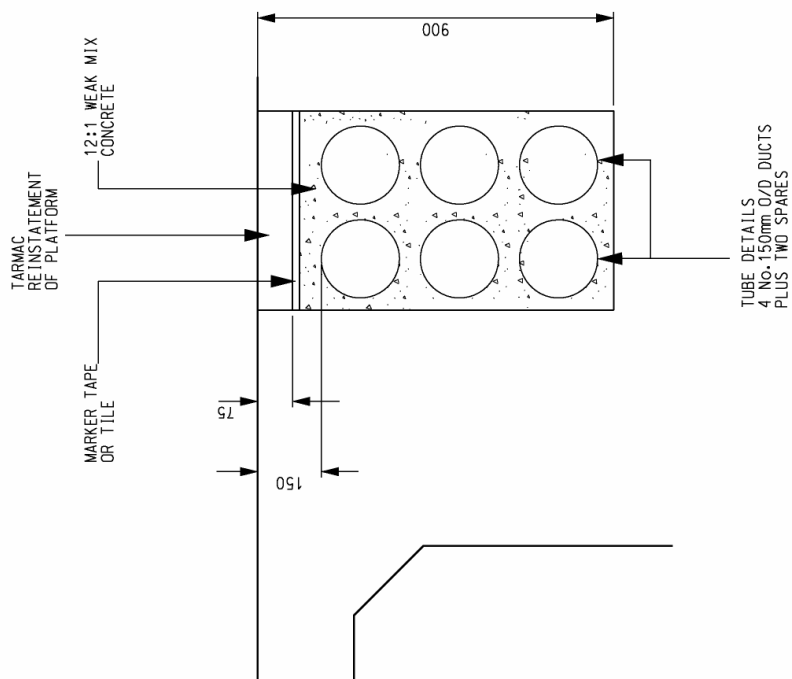
All changes to this CAD generated drawing MUST be made via the master CAD file.

Size: A3

DO NOT SCALE

NOTES

REDRAWN FROM SOUTHERN REGION DRAWING SR/A4/12930
ALL DIMENSIONS, UNLESS INDICATED OTHERWISE,
ARE IN mm



CO1	23/11/04	ISSUED FOR CONSTRUCTION	MA	AA	AA
	Rev.	Date	Description of Revisions		
Project Manager:		Network Rail Project Manager:			Network Rail Application Manager:

CONSTRUCTION ISSUE



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Fax: 0207 620 5632

Project Title

Ref:	NR/SP/ELP/27224
Issue:	2
Date:	April 2006

ARRANGEMENT OF UPVC TUBE ROUTE FOR H.V. & PILOT CABLES THROUGH PLATFORMS

Designed	Checked	T. BENNETT
Approved	Responsible Engineer	
Drawn By		N. JORDAN
Date		15/10/03
Scale		1:10 @ A3
SHEET NO. 1 OF 1		
PROJECT TITLE: DESIGN/SOUTHERN REGION/CIVIL DRAWINGS		
Drawing Number		
A437/00/CV/40558		
Revision		
C02		

Size: A3

All changes to this CAD generated drawing MUST be made via the master CAD file.

DO NOT SCALE

NOTE :

IF A WEARING COURSE IS LAID THE THICKNESS OF THE BASE COURSE MAY BE REDUCED BY THE THICKNESS OF THE WEARING COURSE.
REDRAWN FROM SOUTHERN REGION DRAWING SR/A3/12616

LAYER	PERMITTED MATERIALS	THICKNESS OF LAYERS FOR ROADS				DIAGMMATIC CROSS SECTION OF TRENCH							
		TYPE 1 mm inches	TYPE 11 mm inches	TYPE 111 mm inches	TYPE 1V mm inches								
WEARING COARSE	IF EXISTING SURFACE IS ROLLED ASPHALT :- ROLLED ASPHALT	40 1.5	40 1.5	40 1.5	40 1.5	<table><tr><td>WEARING COURSE</td></tr><tr><td>BASE COURSE</td></tr><tr><td>ROAD BASE</td></tr><tr><td>SUB-BASE</td></tr><tr><td>BACK-FILL</td></tr><tr><td>FINE BACK - FILL OVER APPARATUS</td></tr><tr><td></td></tr></table>	WEARING COURSE	BASE COURSE	ROAD BASE	SUB-BASE	BACK-FILL	FINE BACK - FILL OVER APPARATUS	
	WEARING COURSE												
BASE COURSE													
ROAD BASE													
SUB-BASE													
BACK-FILL													
FINE BACK - FILL OVER APPARATUS													
													
BASE COARSE	IF EXISTING SURFACE IS NOT ROLLED ASPHALT :- FINE COLD ASPHALT OR DENSE BITUMEN	25 1.0	20 .75	20 .75	20 .75								
	IF EXISTING SURFACE IS ROLLED ASPHALT :- DENSE TAR MACADAM OR DENSE BITUMEN MACADAM OR ROLLED ASPHALT	65 2.5	65 2.5	65 2.5	40 1.5								
SEE NOTE	IF EXISTING SURFACE IS NOT ROLLED ASPHALT :- SINGLE COURSE TAR MACADAM TO BS4987 OR SINGLE COURSE BITUMEN MACADAM TO BS 4987	75 3.0	60 2.25	60 2.25	60 2.25								
ROAD BASE	ROLLED ASPHALT BASE OR DENSE TAR OR DENSE BITUMEN BASE OR CEMENT BOUND GRANULAR OR LEAN CONCRETE OR WET MIX MACADAM OR DRY BOUND MACADAM	130 5.0	110 4.5	90 3.5									
		160 6.25	160 6.25	100 4.0									
		Not Permitted		150 6.0	100 4.0								
		Not Permitted	190 7.5	150 6.0	100 4.0								
SUB-BASE	GRANULAR SUB-BASE 40mm. MAX SIZE (TYPE1) OR GRANULAR SUB-BASE (TYPE 11) OR CEMENT BOUND GRANULAR OR LEAN CONCRETE OR SUITABLE EXCAVATED ROAD PAVEMENT	240 9.5	200 8.0	160 6.25	100 4.0								
		280 11.0	260 10.0	210 8.25	150 6.0								

TYPES OF PUBLIC HIGHWAYS

- 1 : TRUNK AND PRINCIPLE DUAL-CARRIAGEWAY ROADS, ABOUT 50% OF WHOSE LENGTH WITHIN THE AREA OF ANY GIVEN HIGHWAY AUTHORITY IS DUAL.
- 11 : ALL THE OTHER 'A' ROADS AND ROADS WITH HOT ROLLED ASPHALT SURFACING (BUT THERE MAY BE OBVIOUS EXCEPTIONS TO THIS CRITERION, FOR EXAMPLE WHEN HOUSING ESTATE ROADS AND CUL-DE-SACS ARE SURFACED WITH HOT ROLLED ASPHALT).
- 111 : ROADS NOT IN TYPES 1 AND 11 WHICH CARRY BUS SERVICES.
- 1V : ALL OTHER PUBLIC HIGHWAYS.

NOTE: THESE DETAILS APPLY ONLY TO PUBLIC HIGHWAYS : NOT PRIVATE ROADS, DRIVEWAYS ETC.



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Network Rail Project Manager

FOR CONSTRUCTION

Client

Ref:	NR/SP/ELP/27224
Issue:	2
Date:	April 2006
Project Title	GUIDE TO TYPES OF FLEXIBLE ROAD CONSTRUCTION FOR REINSTATEMENT OF TRENCHES IN CARRIAGEWAYS
Designated	Checked T. BENNETT
Approved	N. JORDAN
Date	15/10/03
Scale	NTS
CAD Ref.	DESIGN/SOUTHERN REGION/CIVIL DRAWINGS
Drawing Number	A437/00/CV/40559
Revision	C01

All changes to this CAD generated drawing MUST be made via the master CAD file.

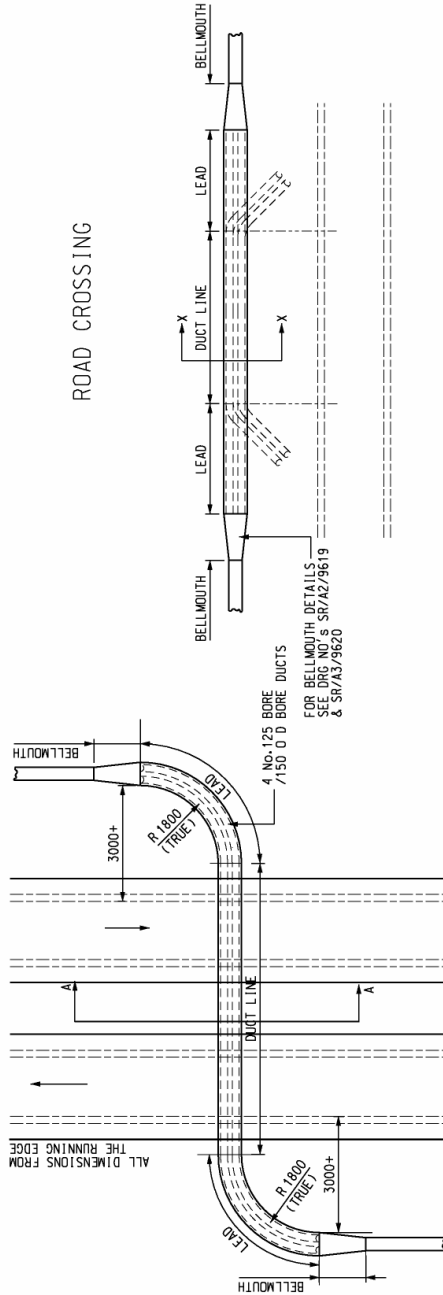
Size: A3

DO NOT SCALE

NOTES:

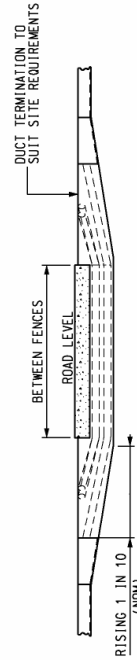
- + DENOTES MINIMUM DIMENSION
- ALL DIMENSIONS, UNLESS INDICATED OTHERWISE, ARE IN mm
- REDRAWN FROM SOUTHERN REGION DRAWING SR/A3/13961

TRACK CROSSING

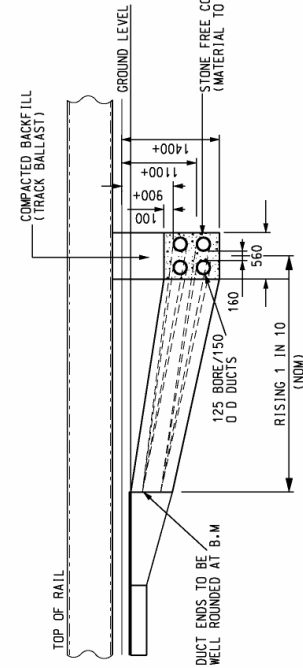


PLAN 1:100

ROAD CROSSING



SECTION X-X



TYPICAL SECTION A-A 1:50



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Network Rail
Network Rail Project Manager

CONSTRUCTION ISSUE

Status

Client

Project Title

Drawing Title

TYPICAL ARRANGEMENT OF
DUCT LINES FOR TRACK
& ROAD CROSSINGS

Designed	Checked	T. BENNETT
Approved	Responsible Engineer	N. JORDAN
Date	Drawn By	03/11/03 K. MEIN
Scale	Sheet	1 of 1
CAD Ref.	.../DESIGN/SOUTHERN REGION/CIVIL DRAWINGS	
Drawing Number	A437/00/CV/40572	

Size: A3

All changes to this CAD generated drawing MUST be made via the master CAD file.

NETWORK RAIL

(Engineering)

BRIEFING NOTE

NR/SP/ELP/27224

Issue 2

**Specification for installation of cable routes forming
part of the traction distribution system**

April 2006

Compliance Date: 1 April 2006

Standard Owner: Head of Electrification & Plant

Business Unit Owner: Engineering

Executive Summary

NR/SP/ELP/27224 was reviewed as part of the SAF3 process and re-issued as a SAF3 Business Process Document. In addition, minor technical changes have been made to the text of the document and the attached drawings to correct conflicts and omissions identified since this standard was first published.

Detailed description of the changes

None

Implementation

The provisions of this document shall be applied from 1 April 2006

Applicability

The provisions of this document are of relevance to the following functions:

Function	Applicable
Contracts & Procurement	
Corporate Change	
Corporate Planning & Regulatory Affairs	
Engineering	√
Finance	
Government & Corporate Affairs	
Human Resources	
Information Management	
Legal Services	
Maintenance	√
Network Development	
Major Projects & Investment	√
Major Projects & Telecoms	
National Delivery Service	
Operations & Customer Services	
Railway Estates	
Safety & Compliance	
Strategic Analysis	

Technical Workscope Applicability

Technical Workscope Applicability (Please tick relevant boxes) Key: M, I & T = Manufacture, Inspection & Test at Works Comm. = Commissioning Inst. = Installation	Design		Equipment			Construction				
	General	System Design	General	Design	M I & T	General	Equipment Site Inst.	Equipment Site Test	System Comm.	Maintenance
A – All										
C&S – Civil & Structural										
E&P – Electrification (AC and/or DC) & Plant	√	√	√	√	√	√	√	√	√	√
M – Maintenance	√	√	√	√	√	√	√	√	√	√
P – Track										
S – Signalling										
T – Telecoms										
H – Safety & Site Working										

Contact Name

Requests for further information or enquiries may be directed to Arif Arabaji, 085 78978.